

Methods to Find Parameters

Analytical

$$\nabla J(\theta) = 0$$

$$\nabla (E^T E) = 0$$

$$\theta^* = (X^T X)^{-1} X^T Y \text{ (Normal Eqn)}$$

Iterative

- Initial guess

- Move in the direction opp. to Gradient

$$\begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \end{bmatrix}^{\text{new}} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \end{bmatrix}^{\text{old}} - \alpha \nabla J(\theta)$$

Gradient Descent on Linear Regression

$$\theta^{\text{new}} = \theta^{\text{old}} - \alpha \nabla J(\theta) \quad J = \sum \epsilon_i^2$$

I

$$J = \sum \epsilon_i^2 = E^T E$$

$$J = (Y - X\theta)^T (Y - X\theta)$$

$$= (Y^T - \theta^T X^T)(Y - X\theta)$$

$$= Y^T Y - Y^T X\theta - \theta^T X^T Y + \theta^T X^T X \theta$$

$$\nabla J(\theta) = 0 - X^T Y - X^T Y + (X^T X + X^T X) \theta$$

$$= -2X^T Y + 2X^T X \theta$$

$$\underline{\underline{\nabla J(\theta) = 2X^T (X\theta - Y)}}$$

$$\theta^{\text{new}} = \theta^{\text{old}} - \alpha (2X^T (X\theta^{\text{old}} - Y))$$

J can be changed according to question.

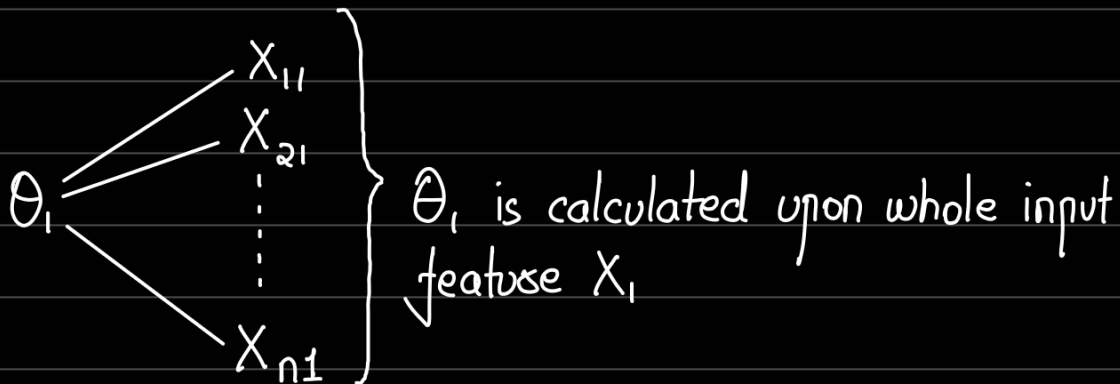
$$\text{II} \quad J = \sum \varepsilon_i^2 = \sum_{i=1}^n (y_i - (\theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2} + \dots + \theta_d x_{id}))^2$$

$$\theta^{\text{new}} = \theta^{\text{old}} - \alpha \nabla J$$

$$\begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_d \end{bmatrix}^{\text{new}} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_d \end{bmatrix}^{\text{old}} - \alpha \begin{bmatrix} \partial J / \partial \theta_0 \\ \partial J / \partial \theta_1 \\ \partial J / \partial \theta_2 \\ \vdots \\ \partial J / \partial \theta_d \end{bmatrix}$$

$(d+1) \times 1$
 $(d+1) \times 1$

So taking θ_1 as example:



$$\theta_1^{\text{new}} = \theta_1^{\text{old}} - \alpha \frac{\partial J}{\partial \theta_1}$$

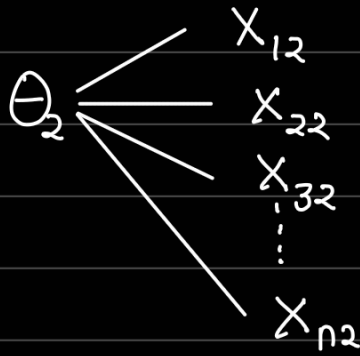
$$\theta_1^{\text{new}} = \theta_1^{\text{old}} - \alpha \sum_{i=1}^n (2 \varepsilon_i) (-x_{i1})$$

$y_i - (\theta_0 + \theta_1 x_{i1} + \dots + \theta_d x_{id})$

Coefficient of θ_1
 (-ve sign due to)

$y_i - (\theta_0 + \theta_1 x_{i1} + \dots)$

Similarly, for θ_2 :



$$\theta_2^{\text{new}} = \theta_2^{\text{old}} - \alpha \frac{\partial J}{\partial \theta_2}$$

$$\theta_2^{\text{new}} = \theta_2^{\text{old}} - \alpha \sum_{i=1}^n (2\varepsilon_i)(-x_{i2})$$

So, for any θ_j :

$$\theta_j^{\text{new}} = \theta_j^{\text{old}} - \alpha \frac{\partial J}{\partial \theta_j}$$

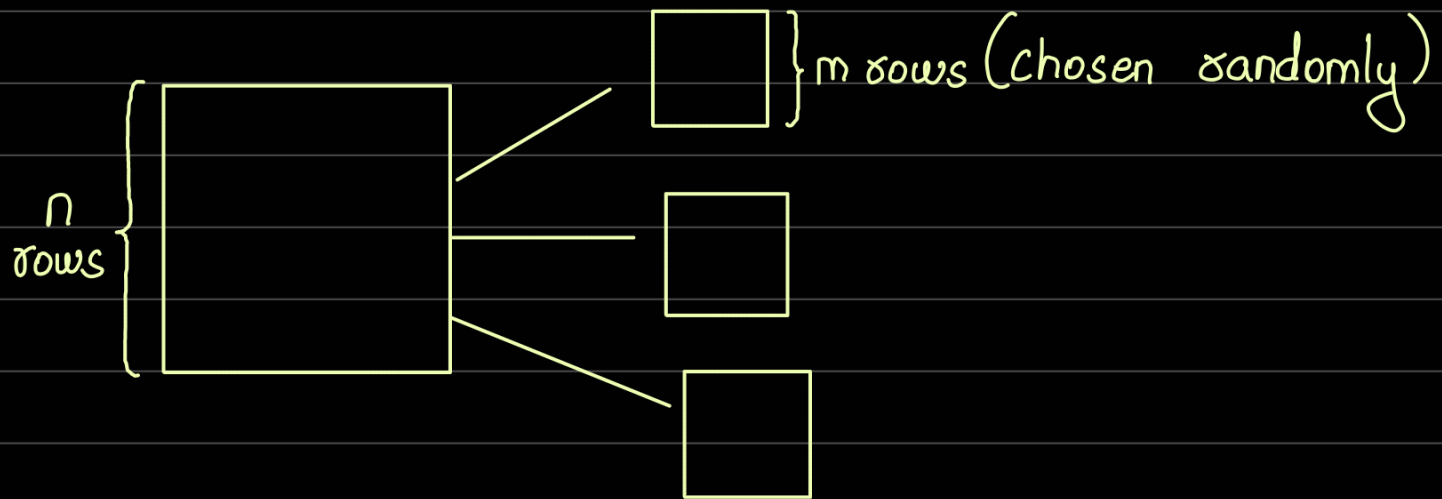
$$= \theta_j^{\text{old}} - \alpha \sum_{i=1}^n (2\varepsilon_i)(-x_{ij})$$

Batch Gradient Descent

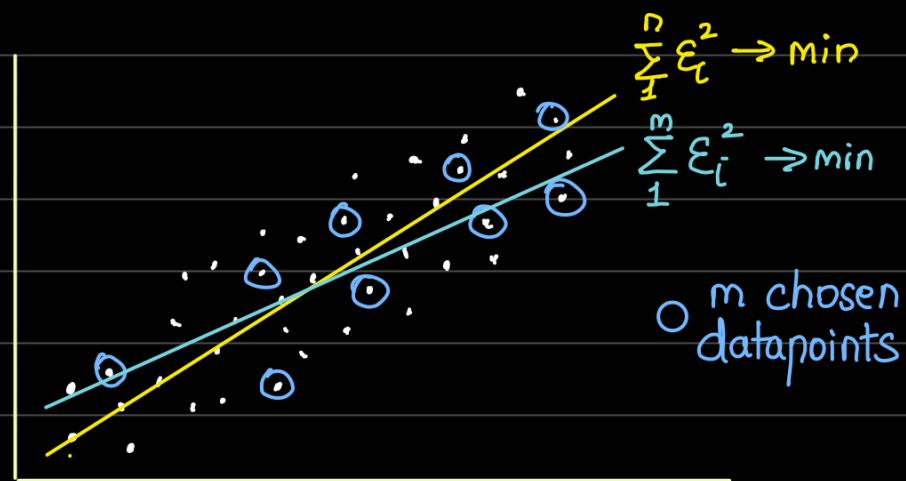
Batch Gradient Descent uses the whole training set to compute the gradient at every step, which makes it very slow when the training set is large. Like we did above.

In linear regression, BGD with an optimal learning rate can converge to the optimal solution.

Mini Batch Gradient Descent

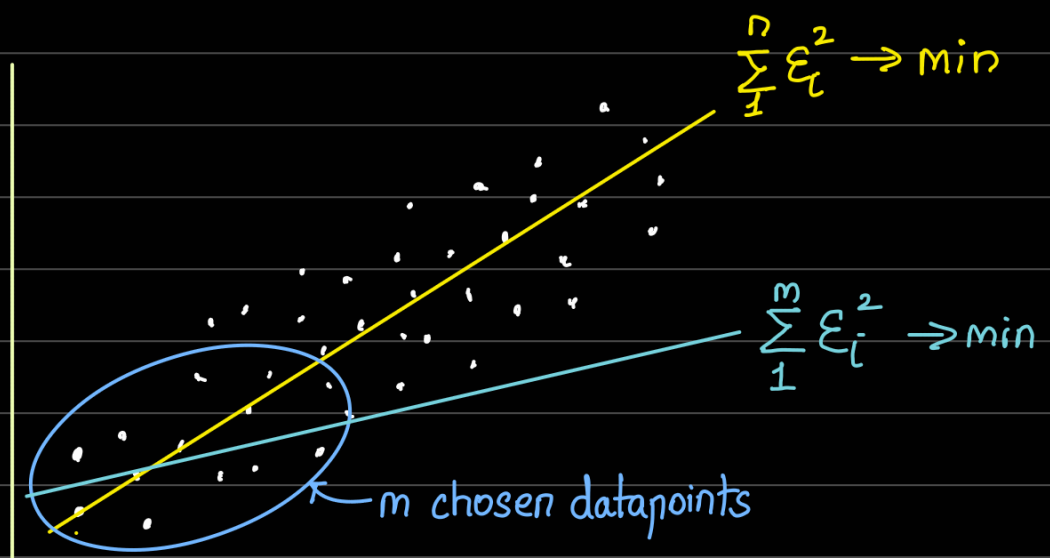


Now Gradient is, $J = \sum_{i=1}^m \mathcal{E}_i^2$

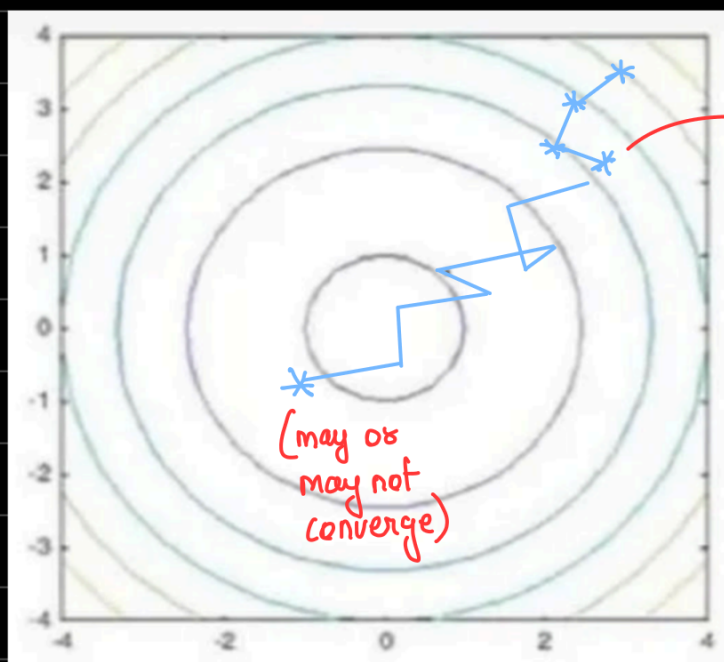


The level curve / contour map of both $\sum_{i=1}^n \mathcal{E}_i^2$ and $\sum_{i=1}^m \mathcal{E}_i^2$ can be similar but will not be same.

For eg $\rightarrow$



So, we can say from figure above that error is increased with respect to error with n data points.



Error increased

(may or may not converge)

Here, we are not optimizing $\sum_{i=1}^n \epsilon_i^2$ — Goal (to converge)

but we are trying to optimize $\sum_{i=1}^m \epsilon_i^2$ — Goal (to optimize according to m datapoints)

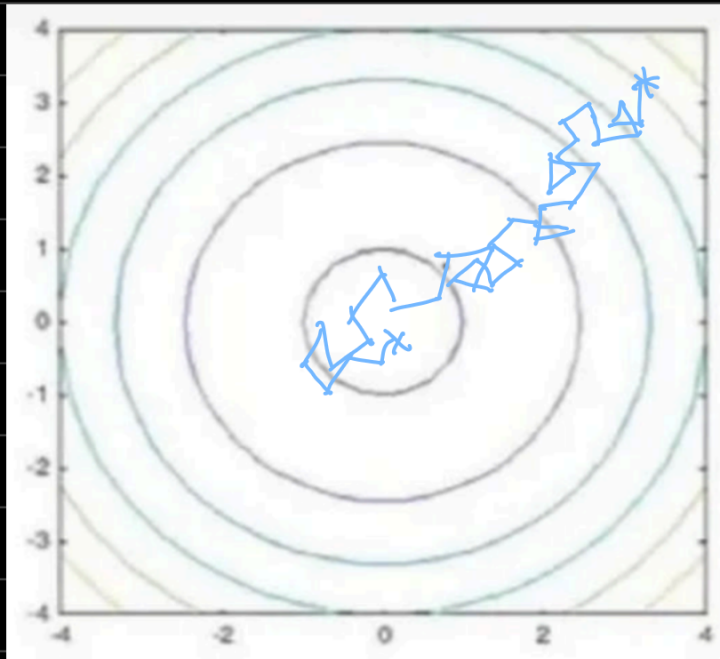
So, To reach $\nabla J = 0$ is target of $\sum_{i=1}^n \epsilon_i^2$, but not the target of $\sum_{i=1}^m \epsilon_i^2$, if we reach $\nabla J = 0$ using $\sum_{i=1}^m \epsilon_i^2$, it's a coincidence.

* Mini BGD updates model parameters using small subset of the training data

* This do not converge on optimal solution.

Stochastic Gradient Descent

* In SGD we take 1 row at a time and will increase noise.



For example, we need to update $\theta_j \rightarrow$

$$\theta_j^{\text{new}} = \theta_j^{\text{old}} - \alpha (2\varepsilon_i - x_{ij}) \quad \begin{matrix} \{ \text{fast} \} \\ \swarrow \\ \text{chosen point (Single)} \end{matrix}$$

Fast $\rightarrow$ SGD $>$ MBGD $>$ BGD
Stable $\rightarrow$ SGD $<$ MBGD $<$ BGD

- SGD updates the model parameters after computing the gradient of the cost function with respect to only one randomly chosen training example at a time.
- The randomness in SGD introduces noise into the parameter updates, which helps the algorithm to escape local minima more effectively than BGD

- When the cost function is very irregular SGD has a better chance of finding global minimum than BGD.
- SGD do not converge on optimum solution.

Batch/Vanilla Gradient Descent(BGD/VGD)

- 1 Use the entire training set to compute the gradient at each epoch.
- 2 Makes exact estimate for $\nabla_{\theta} L(\theta)$
- 3 Very slow when the training set is large.
- 4 In linear regression, BGD with optimal learning rate, is guaranteed to converge to the optimal solution.

Mini-Batch Gradient Descent(MGD)

- 1 $L(\theta) = \sum_{i=1}^k \mathcal{E}_i^2$
- 2 Do parameter updates for batches of points each of size k .
- 3 Higher the k , better the estimates of $\nabla_{\theta} L(\theta)$
- 4 MGD is not guaranteed to converge to the optimal solution.

Stochastic Gradient Descent(SGD)

- 1 $L(\theta) = \mathcal{E}_i^2$ Update parameters for each random data point i .
- 2 "Stochastic" because we are estimating the total gradient based on a single data point.
- 3 A parameter update which is locally favourable to one point may harm other points. (**GREEDY**)
- 4 The randomness in SGD helps the algorithm escape local minima effectively.
- 5 When $L(\theta)$ is very irregular SGD has a better chance of converging compared to VGD.
- 6 SGD is not guaranteed to converge to the optimum solution.

Q Which of the following statements are correct?

- 1 We can get multiple local optimum solutions if we solve a linear regression problem by minimizing the sum of squared errors using gradient descent.
- 2 For convex loss functions (i.e., with a bowl shape), stochastic gradient descent is guaranteed to eventually converge to the global optimum.
- 3 For convex loss functions (i.e., with a bowl shape), batch gradient descent is guaranteed to eventually converge to the global optimum while stochastic gradient descent is not.

Options:

- A. (1, 2)
- B. (1, 3)
- C. (1)
- D. (3)

(D) Ans

Which of the following is a true statement about gradient descent (GD)?

Options:

- A) Gradient descent can be slow to run on datasets with a large number of samples.
- B) When training, we should not update the bias (intercept) term using GD.
- C) Reducing the learning rate always guarantee a reduction in error.
- D) Gradient descent requires only one update step to converge, unlike stochastic gradient descent (SGD).

(A) Ans.

Which of the following statements about gradient descent (GD) and stochastic gradient descent (SGD) are *true*?

Options:

- A) Both GD and SGD provide unbiased estimates of the true gradient at each step.
- B) The memory and compute requirements of a single update step for both methods scale linearly with the number of features.
- C) The memory and compute requirements of a single update step for GD scale with the number of data points, but for SGD, they do not.
- D) GD generally requires fewer iterations to converge than SGD, given a well-chosen learning rate.

(B, C, D) Ans.

1 Gradient descent is used in modeling to:

- ☐ compute the loss.
- ☐ compute the gradient of the loss.
- ☐ maximize the loss.
- ☒ minimize the loss.

2 For a model with k parameters and a dataset with n data points in d dimensions, how many dimensions is the gradient of the loss?

- ☐ 1
- ☒ k
- ☐ $k + 1$
- ☐ d
- ☐ n

3 Each iteration of stochastic gradient descent:

- ☐ strictly increases the loss.
- ☐ strictly decreases the loss.
- ☒ may increase or decrease the loss.

4 The gradient of the loss in stochastic gradient descent is:

- ☐ always negative (all entries in the gradient are negative).
- ☐ always equal to zero (all entries in the gradient are 0).
- ☒ computed on a sample or subset of the dataset.
- ☐ computed on the entire dataset.

1 Which of the following settings would most justify the use of stochastic gradient descent instead of gradient descent?

- ☐ All entries in the gradient of the loss are negative.
- ☒ The dataset is large.
- ☐ The data was constructed from a simple random sample.
- ☐ The loss function is not defined.

2 Which of the following could affect whether or not gradient descent converges to the global minimum?

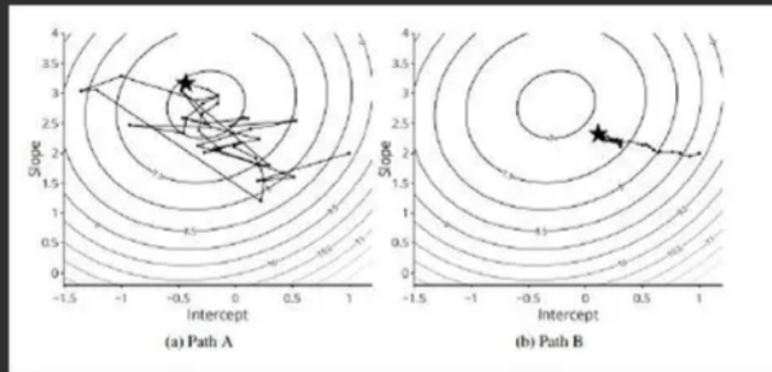
(a) Learning Rate

- ☒ True
- ☐ False

(b) Initialization of parameters

- ☒ True
- ☐ False

The following figures depict the contours of the loss surface as well as the solution path of parameters obtained by running an optimization algorithm. The star corresponds to the final parameter value. Use the figures to answer the following questions:



- (a) How many parameters does this model have?
☐ 1 ☐ 2 ☐ 3 ☐ 4
- (b) The sequence of parameters in Path A more likely came from an execution of:
☐ Gradient Descent ☐ Stochastic Gradient Descent
- (c) Which optimization path got closer to the optimal solution?
☐ Path A ☐ Path B

- (a) 2 (Slope, intercept)
- (b) Stochastic Gradient Descent
- (c) Path A

Which of the following is true, given the optimal learning rate?

- i. Batch gradient descent is always guaranteed to converge to the global optimum of a loss function.
- ii. Stochastic gradient descent is always guaranteed to converge to the global optimum of a loss function.
- iii. For convex loss functions (i.e., with a bowl shape), batch gradient descent is guaranteed to eventually converge to the global optimum while stochastic gradient descent is not.
- iv. For convex loss functions (i.e., with a bowl shape), stochastic gradient descent is guaranteed to eventually converge to the global optimum while batch gradient descent is not.
- v. For convex loss functions (i.e., with a bowl shape), both stochastic gradient descent and batch gradient descent will eventually converge to the global optimum.
- vi. For convex loss functions (i.e., with a bowl shape), neither stochastic gradient descent nor batch gradient descent are guaranteed to converge to the global optimum.

(iii) Ans.